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Course Code : MCS-201

Course Title : Programming in C and PYTHON

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Maximum Marks : 100

Weightage : 30% www.ignousite.com

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SECTION-A (C-Programming)

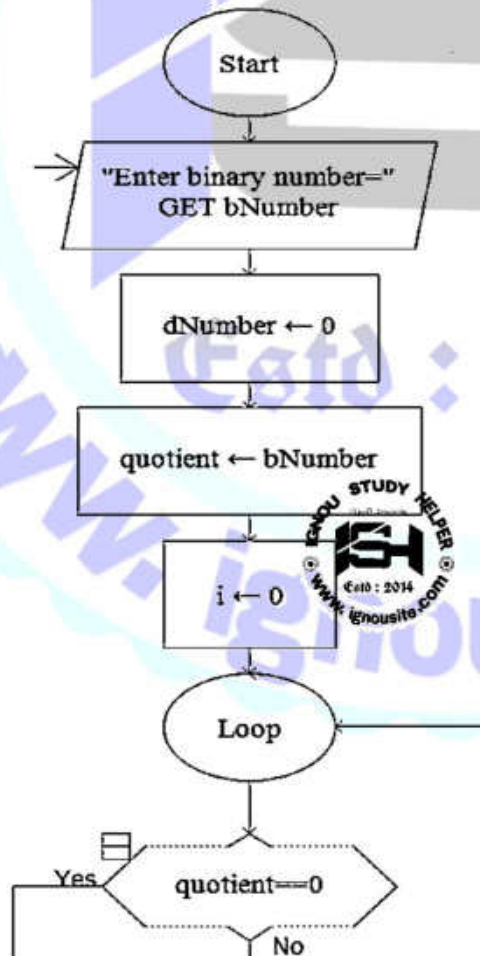
Question1: Write an algorithm, draw a flow chart and write its corresponding C program to convert a Binary number to its equivalent Decimal number.

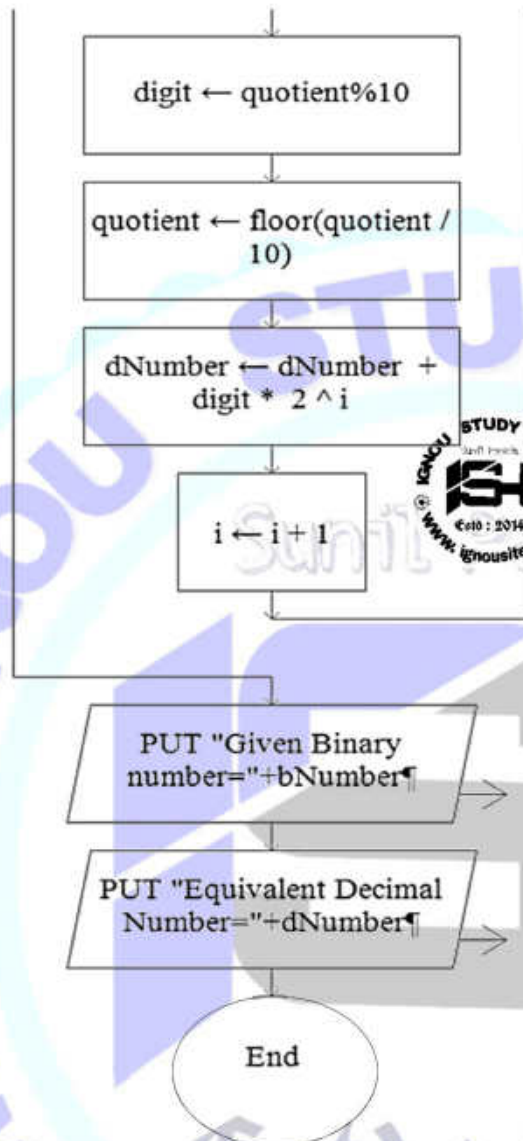
Ans.

An algorithm Binary number to Decimal number:

- Take a binary number as the input.
- Divide the number by 10 and store the remainder into variable rem.
- $\text{decimal_num} = \text{decimal_num} + \text{rem} * \text{base}$;
Initially, the decimal_num is 0, and the base is 1, where the rem variable stores the remainder of the number.
- Divide the quotient of the original number by 10.
- Multiply the base by 2.
- Print the decimal of the binary number.

Flow chart:



**Program:**

```

#include <stdio.h>
#include <conio.h>
void main()
{
    // declaration of variables
    int num, binary_num, decimal_num = 0, base = 1, rem;
    printf (" Enter a binary number with the combination of 0s and 1s \n");
    scanf ("%d", &num); // accept the binary number (0s and 1s)
    binary_num = num; // assign the binary number to the binary_num variable
    while ( num > 0)
    {
        rem = num % 10; /* divide the binary number by 10 and store the remainder in rem variable. */
        decimal_num = decimal_num + rem * base;
        num = num / 10; // divide the number with quotient
    }
  
```

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```
base = base * 2;
}
printf ( " The binary number is %d \t", binary_num); // print the binary number
printf ( " \n The decimal number is %d \t", decimal_num); // print the decimal
getch();
}
```

Output

Enter a binary number with the combination of 0s and 1s

1101

The binary number is 1101

The decimal number is 13



Question 2: Write an algorithm and flowchart to calculate the roots of quadratic equation $AX^2 + BX + C = 0$. Transform our algorithm in to an equivalent C program.

Ans.

Algorithm:

Step-1 Start

Step-2 Input A,B,C

Step-3 $DISC = B^2 - 4 A * C$

Step-4 IF (DISC < 0) THEN

Write Roots are Imaginary

Stop

ENDIF

Step-5 IF (DISC == 0) THEN

Write Roots are Real and Equal

$X1 = -B / (2 * A)$

Write Roots are X1,X1

Stop

ENDIF

Step-6 IF (DISC > 0)

Write Roots are Real and Unequal

$X1 = (-B + \text{SQRT}(\text{DISC})) / (2 * A)$

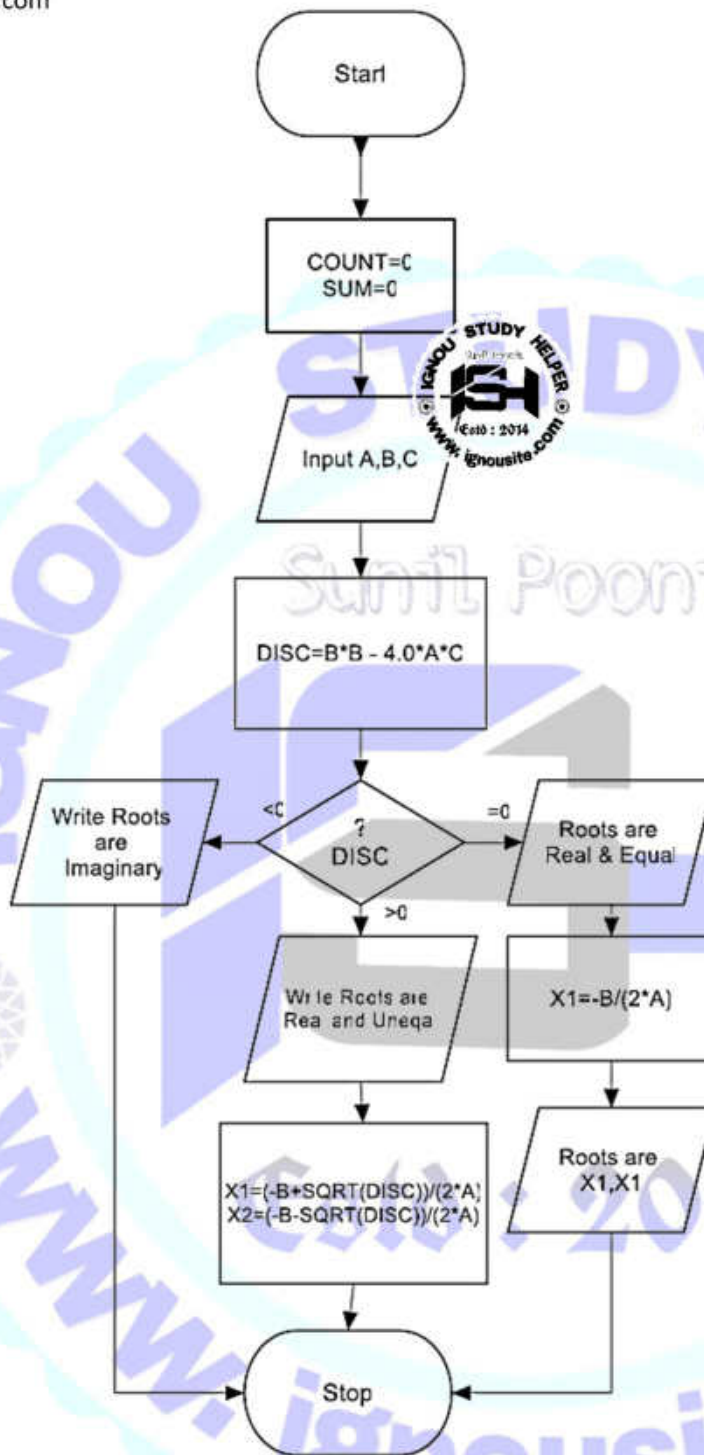
$X2 = (-B - \text{SQRT}(\text{DISC})) / (2 * A)$

Write Roots are X1,X2

Stop

ENDIF

Flow Chart:



Programme:

```

#include <math.h>
#include <stdio.h>
int main() {
    double a, b, c, discriminant, root1, root2, realPart, imagPart;
    printf("Enter coefficients a, b and c: ");
    scanf("%lf %lf %lf", &a, &b, &c);
    
```

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discriminant = $b * b - 4 * a * c$;

// condition for real and different roots

if (discriminant > 0) {

root1 = $(-b + \sqrt{\text{discriminant}}) / (2 * a)$;

root2 = $(-b - \sqrt{\text{discriminant}}) / (2 * a)$;

printf("root1 = %.2lf and root2 = %.2lf", root1, root2);

}

// condition for real and equal roots

else if (discriminant == 0) {

root1 = root2 = $-b / (2 * a)$;

printf("root1 = root2 = %.2lf;", root1);

}

// if roots are not real

else {

realPart = $-b / (2 * a)$;

imagPart = $\sqrt{-\text{discriminant}} / (2 * a)$;

printf("root1 = %.2lf+%.2lfi and root2 = %.2lf-%.2fi", realPart, imagPart, realPart, imagPart);

}

return 0;

}

Output

Enter coefficients a, b and c: 2.3

4

5.6

root1 = -0.87+1.30i and root2 = -0.87-1.30i

Question 3: Write a C program to generate the following pattern:

```
*
**
***
**
*
```

Ans.

```
#include<stdio.h>
```

```
#include<conio.h>
```

```
int main()
```

```
{
```

```
int n, i, j;
```

```
printf("Enter number of rows: ");
```

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```
scanf("%d",&n);
for(i = 1; i <= n; i++)
{
    for(j = 1; j <= i; j++)
    {
        printf("*");
    }
    printf("\n");
}
for(i = n; i > 1; i--)
{
    for(j = 1; j < i; j++)
    {
        printf(" ");
    }
    // ending line after each row
    printf("\n");
}
return 0;
}
```



Output:

Enter number of rows: 3

```
*
* *
* * *
* *
*
```



Question 4: Write a C program to perform the following operation on matrices of dimension (3 X 3)

(a) Addition (b) Multiplication

Ans.

```
#include<stdio.h>
void main()
{
    int i, j, k, a[3][3], b[3][3], c[3][3], d[3][3];
    // Input Matrix A
    printf("\n Enter the element of matrix A");
    for(i=0;i<3;i++)
    {
        for(j=0;j<3;j++)
        {
            printf("\n Enter the value of A[%d][%d] ",i,j);
            scanf("%d",&a[i][j]);
        }
    }
}
```


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```
}  
}  
// Input Matrix B  
printf("\n Enter the element of matrix B");  
for(i=0;i<3;i++)  
{  
for(j=0;j<3;j++)  
{  
printf("\n Enter the value of B[%d][%d] ",i,j);  
scanf("%d",&b[i][j]);  
}  
}  
// Input Matrix C  
printf("\n Enter the element of matrix C");  
for(i=0;i<3;i++)  
{  
for(j=0;j<3;j++)  
{  
printf("\n Enter the value of C[%d][%d] ",i,j);  
scanf("%d",&c[i][j]);  
}  
}  
// Initialize Matrix D  
d[i][j]=0;  
}  
// Matrix Multilication  
for(i=0;i<3;i++)  
{  
for(j=0;j<3;j++)  
{  
for(k=0;k<3;k++)  
{  
d[i][j]=(d[i][j] + (b[i][k] * c[k][j]));  
}  
}  
}  
// Matrix Addition  
for(i=0;i<3;i++)  
{  
for(j=0;j<3;j++)  
{  
d[i][j]=d[i][j]+a[i][j];  
}  
}  
// Print Matrix D  
for(i=0;i<3;i++)
```

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```
{  
for(j=0;j<3;j++)  
{  
printf("\t%d",d[i][j]);  
}  
printf("\n");  
}  
}
```

Output:

Enter the Element of First Matrix

Enter the value of A[0][0] : 6

Enter the value of A[0][1] : 9

Enter the value of A[0][2] : 1

Enter the value of A[1][0] : 7

Enter the value of A[1][1] : 2

Enter the value of A[1][2] : 4

Enter the value of A[2][0] : 8

Enter the value of A[2][1] : 3

Enter the value of A[2][2] : 7

Enter the Element of Second Matrix

Enter the value of B[0][0] : 1

Enter the value of B[0][1] : 4

Enter the value of B[0][2] : 2

Enter the value of B[1][0] : 3

Enter the value of B[1][1] : 2

Enter the value of B[1][2] : 5

Enter the value of B[2][0] : 4

Enter the value of B[2][1] : 6

Enter the value of B[2][2] : 3

Enter the Element of Third Matrix

Enter the value of C[0][0] : 2



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Enter the value of C[0][1] : 6

Enter the value of C[0][2] : 3

Enter the value of C[1][0] : 8

Enter the value of C[1][1] : 1

Enter the value of C[1][2] : 5

Enter the value of C[2][0] : 6

Enter the value of C[2][1] : 2

Enter the value of C[2][2] : 4

52 23 32

59 32 43

82 39 61

Question 5: Write a C program to take a list of N numbers, separate even and odd numbers and put them in two appropriate files (evenfile and oddfile). Use File Handling concept.

Ans.

```
#include<stdio.h>
#include
void main()
{
FILE *even, *odd;
int n, array[100], i;
printf("\n How many numbers you want to take... ");
scanf("%d",&n);
for(i=0;i<n;i++)
{
printf("Enter the number... ");
scanf("%d",&array[i]);
}
even=fopen("evenfile","w");
odd=fopen("oddfile","w");
for(i=0;i<n;i++)
{
if(array[i]%2 == 0)
putw(array[i],even);
else
putw(array[i],odd);
}
fclose(even);
fclose(odd);
}
```

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Output:

How many numbers you want to take...

Question 6: Write a program to enter a list of strings and rearrange them in alphabetical order, using a one-dimensional array of pointers, where each pointer indicates the beginning of a string.

Ans.

```
#include<stdio.h>
#include<conio.h>
#include<string.h>
void main()
{
    char *x[20];
    int i,n=0;
    void reorder(int n,char *x[]);
    clrscr();
    printf("Enter no. of String : ");
    scanf("%d",&n);
    printf("\n");

    for(i=0;i<n;i++)
    {
        printf("Enter the Strings %d : ",i+1);
        x[i]=(char *)malloc(20*sizeof(char));
        scanf("%s",x[i]);
    }

    reorder(n,x);
    printf("\nreorder list is : \n");
    for(i=0;i<n;i++)
    {
        printf("%d %s\n",i+1,x[i]);
    }
    getch();
}

void reorder(int n,char *x[])
{
    int i,j;
    char t[20];
    for(i=0;i<n-1;i++)
        for(j=i+1;j<n;j++)
            if(strcmp(x[i],x[j])>0)
            {
                strcpy(t,x[j]);
                strcpy(x[j],x[i]);
                strcpy(x[i],t);
            }
}
```

```
    }  
    return;
```

```
}
```

Output



```
Command Prompt - tc  
Enter no. of String : 5  
Enter the Strings 1 : abh  
Enter the Strings 2 : abc  
Enter the Strings 3 : adh  
Enter the Strings 4 : aaa  
Enter the Strings 5 : abj  
  
reorder list is :  
1 aaa  
2 abc  
3 abh  
4 abj  
5 adh  
_
```

Question 7: Write a C program to sort a list of N numbers

Ans.

```
#include <stdio.h>  
#define MAXSIZE 10
```

```
void main()
```

```
{  
    int array[MAXSIZE];  
    int i, j, num, temp;  
  
    printf("Enter the value of num \n");  
    scanf("%d", &num);  
    printf("Enter the elements one by one \n");  
    for (i = 0; i < num; i++)  
    {  
        scanf("%d", &array[i]);  
    }  
    printf("Input array is \n");  
    for (i = 0; i < num; i++)  
    {  
        printf("%d\n", array[i]);  
    }  
}
```




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```
/* Bubble sorting begins */  
for (i = 0; i < num; i++)  
{  
    for (j = 0; j < (num - i - 1); j++)  
    {  
        if (array[j] > array[j + 1])  
        {  
            temp = array[j];  
            array[j] = array[j + 1];  
            array[j + 1] = temp;  
        }  
    }  
}  
printf("Sorted array is...\n");  
for (i = 0; i < num; i++)  
{  
    printf("%d\n", array[i]);  
}  
}
```

Output:

Enter the value of num

6

Enter the elements one by one

23

45

67

89

12

34

Input array is

23

45

67

89

12

34

Sorted array is...

12

23

34

45

67

89

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Question 8: Write a C program to print reverse of a string (without using strrev() function), and checks that the entered string is a palindrome or not

Ans.

```
#include <string.h>
#include <conio.h>
#include <stdio.h>

void main()
{
    int i,j,len=0;
    char str[50],revstr[50];
    //clrscr();
    printf("\n Enter a String to Reverse : ");
    gets(str);
    for(i=0; str[i]!='\0'; i++)
    {
        len++;
    }

    j=0;
    for(i=len-1; i>=0; i--)
    {
        revstr[j++]=str[i];
    }
    printf("\n Reverse of the Given String is: %s",revstr);
    free(str);
    getch();
}
```

Output

Enter a String to Reverse : www.ignousite.com

Reverse of the Given String is: moc.etsuongi.www

Checks that the entered string is a palindrome or not:

```
#include <stdio.h>
#include <string.h>
int main(){
    char string1[20];
    int i, length;
    int flag = 0;

    printf("Enter a string:");
    scanf("%s", string1);
```

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```
length = strlen(string1);

for(i=0;i < length ;i++){
    if(string1[i] != string1[length-i-1]){
        flag = 1;
        break;
    }
}

if (flag) {
    printf("%s is not a palindrome", string1);
}
else {
    printf("%s is a palindrome", string1);
}
return 0;
}
```



Output

```
Enter a string: www.ignousite.com
www.ignousite.com is not a palindrome
```

SECTION-B (PYTHON-Programming)

Question 9: Write a program in Python Program to display the Fibonacci Sequences up to nth term where n is provided by the user

Ans.

```
# Program to display the Fibonacci sequence up to n-th term where n is provided by the user
# take input from the user
nterms = int(input("How many terms? "))
# first two terms
n1 = 0
n2 = 1
count = 2
# check if the number of terms is valid
if nterms <= 0:
    print("Plese enter a positive integer")
elif nterms == 1:
    print("Fibonacci sequence:")
    print(n1)
else:
    print("Fibonacci sequence:")
    print(n1,"",n2,end=', ')
    while count < nterms:
        nth = n1 + n2
```




```
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print(nth,end=' ')
# update values
n1 = n2
n2 = nth
count += 1
```

Output :



```
How many terms? 10
Fibonacci sequence:
0, 1, 1, 2, 3, 5, 8, 13, 21, 34,
```

Question 10: Write a Program in Python that Accepts a Sentence and Calculate the Number of Digits, Uppercase and Lowercase Letters.

Ans.

```
def string_test(s):
    d={"UPPER_CASE":0, "LOWER_CASE":0}
    for c in s:
        if c.isupper():
            d["UPPER_CASE"]+=1
        elif c.islower():
            d["LOWER_CASE"]+=1
        else:
            pass
    print ("Original String : ", s)
    print ("No. of Upper case characters : ", d["UPPER_CASE"])
    print ("No. of Lower case Characters : ", d["LOWER_CASE"])
```

```
string_test('The quick Brown Fox')
```

Output:

```
Original String : The quick Brow Fox
No. of Upper case characters : 3
No. of Lower case Characters : 13
```



Question 11: Create a module series.py containing functions to determine Fibonacci series and Exponential series. Import the module created to make it accessible, and Call the functions of that module with module name . Demonstrate the access of functions in the module created.

Ans.

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creating module name series.py

```
def factorial(n):                # function to calculate factorial
    fact=1
    for i in range(1,n+1):
        fact=fact*i
    return fact

def fibonacci(n):                # function to display fibonacci series
    first=0
    second=1
    print("FIBONACCI SERIES:\n",first," ",second,end='')
    for i in range(1,n-1):
        third=first+second
        print(" ",third,end='')
        first=second
        second=third

def exponential(x,n):            # function to display exponential series upto n
    s=0
    for i in range(n):
        print("{x pow",i,"/",i,"!")+"",end='')
        s=s+(x**i/factorial(i))
    print("\nExponential series sum:",s )

import series # importing module series.py

print ("Here we are using module series.py")

n=int(input("enter number of terms in fibonacci series "))

series.fibonacci(n) # calling a function present in other module
```

Output:

```
Here we are using module series.py
enter number of terms in fibonacci series 10
FIBONACCI SERIES:
0 1 1 2 3 5 8 13 21 34
```

Question 12: Differentiate Between Modules and Scripts, give suitable python code for each.

Ans.

Modules: A Python module is a file containing Python definitions and statements. A module can define functions, classes, and variables. A module can also include runnable code. Grouping related code into a module makes the code easier to understand and use. It also makes the code logically organized.

Code:

```
# importing sqrt() and factorial from the
# module math
from math import sqrt, factorial
```



```
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# if we simply do "import math", then
# math.sqrt(16) and math.factorial()
# are required.
print(sqrt(16))
print(factorial(6))
```

Output:

4.0
720



Script is a collection of commands in a file designed to be executed like a program. The file can of course contain functions and import various modules, but the idea is that it will be run or executed from the command line or from within a Python interactive shell to perform a specific task. Often a script first contains a set of function definitions and then has the main program that might call the functions.

Consider this script, found in \$UWHPSC/python/script1.py:

Code:

```
# Declare a fruit list
fruits = ["Mango", "Orange", "Guava", "Banana"]

# Insert an item in the 2nd position
fruits.insert(1, "Grape")

# Displaying list after inserting
print("The fruit list after insert:")
print(fruits)

# Remove an item
fruits.remove("Guava")

# Print the list after delete
print("The fruit list after delete:")
print(fruits)
```

Output:

Mango, Orange, Guava, Banana
The fruit list after insert:
Grape, Mango, Orange, Guava, Banana
The fruit list after delete:
Grape, Mango, Orange, Banana

Question 13: Differentiate Between Co-routines and Sub-routines, give suitable python code for each.

Ans.

Coroutines	Subroutines
Unlike subroutines, Coroutines have multiple entry points.	Subroutines have only a single entry point, i.e., the main function.



We can only suspend and resume the execution of Python subroutines from a single point, and it can only be resumed only from the starting point.	We can suspend and resume the execution of coroutines in Python from multiple points in the function, and also, we can resume execution from where we have suspended it last.
Unlike subroutines, we don't have any main function for Python coroutines to order and coordinate their execution particularly.	In Python subroutines, the main function coordinates between the multiple subroutines and controls their executions.
Coroutines are corporative as they together form a pipeline structure while we are executing them.	Subroutines form a linear execution structure.
In Coroutines, we have a coroutine that displays the result for input data.	For subroutines, the processing result of data given in them is displayed by the main function.



Co-routines Code:

```
# Default function to search prefix
def print_name(prfx):
    print("Coroutine object searching for the prefix: {}".format(prfx)) # Searching for prefix
    while True:
        GivenName = (yield)
        if prfx in GivenName: # If required prefix match
            print(GivenName) # Print given name
CorouteObject = print_name("Officer") # Taking prefix = Officer for coroutine object
# Taking names as input from user
Name1 = input("Enter first name: ")
Name2 = input("Enter second name: ")
Name3 = input("Enter third name: ")
CorouteObject.__next__() # using __next__() method to call coroutine
# sending input data to coroutine with send() method
CorouteObject.send(Name1)
CorouteObject.send(Name2)
CorouteObject.send(Name3)
```

Output:

```
Enter first name: Alex
Enter second name: Officer Steve Rogers
Enter third name: Officer Natasha Widow
Coroutine object searching for the prefix: Officer
Officer Steve Rogers
Officer Natasha Widow
```



Sub-routines Code:

```
Def addNum (num1, num2):
    Total = num1 + num2
    Print (total)
```

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```
Num1 = int(input("Please enter a number to be added: "))
```

```
Num2 = int(input("Please enter another number to be added: "))
```

```
addNum (num1, num2)
```

Output:

```
Please enter a number to be added: 10
```

```
Please enter another number to be added: 20
```

```
30
```



Question 14: Write Short notes on Generators, Iterators and Decorators give suitable python code for each.

Ans.

Generators: In this section we learn about Python generators. They were introduced in Python 2.3. It is an easier way to create iterators using a keyword yield from a function.

Code:

```
>>> def my_generator():
...     print("Inside my generator")
...     yield 'a'
...     yield 'b'
...     yield 'c'
...
>>> my_generator()
<generator object my_generator at 0x7fbcfa0a6aa0>
```

Iterators: Python iterator objects are required to support two methods while following the iterator protocol.

`__iter__` returns the iterator object itself. This is used in for and in statements.

`__next__` method returns the next value from the iterator. If there is no more items to return then it should raise StopIteration exception.

Code:

```
class Counter(object):
    def __init__(self, low, high):
        self.current = low
        self.high = high

    def __iter__(self):
        'Returns itself as an iterator object'
        return self

    def __next__(self):
        'Returns the next value till current is lower than high'
        if self.current > self.high:
            raise StopIteration
        else:
            self.current += 1
            return self.current - 1
```




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Decorators: Decorator is way to dynamically add some new behavior to some objects. We achieve the same in Python by using closures.

In the example we will create a simple example which will print some statement before and after the execution of a function.

Code:

```
>>> def my_decorator(func):
...     def wrapper(*args, **kwargs):
...         print("Before call")
...         result = func(*args, **kwargs)
...         print("After call")
...         return result
...     return wrapper
...
>>> @my_decorator
... def add(a, b):
...     "Our add function"
...     return a + b
...
>>> add(1, 3)
Before call
After call
4
```

Question 15: Write a Program in Python to check if a given year is a leap year.

Ans.

Default function to implement conditions to check leap year

```
def CheckLeap(Year):
    # Checking if the given year is leap year
    if((Year % 400 == 0) or
       (Year % 100 != 0) and
       (Year % 4 == 0)):
        print("Given Year is a leap Year");
    # Else it is not a leap year
    else:
        print ("Given Year is not a leap Year")
# Taking an input year from user
Year = int(input("Enter the number: "))
# Printing result
CheckLeap(Year)
```

Output:

```
Enter the number: 2004
Given year is a leap Year
```


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Question 16: Briefly discuss the Lambda, map() and filter() function; with suitable code for each.

Ans. Python Lambda Functions are anonymous function means that the function is without a name. As we already know that the def keyword is used to define a normal function in Python. Similarly, the lambda keyword is used to define an anonymous function in Python.

Code:

```
# Python program to demonstrate  
# lambda functions
```

```
x="IgnouStudyHelper"
```

```
# lambda gets pass to print  
(lambda x : print(x))(x)
```

Output:

IgnouStudyHelper

The map() function is a higher-order function. As previously stated, this function accepts another function and a sequence of 'iterables' as parameters and provides output after applying the function to each iterable in the sequence.

Code:

```
def function(a):  
    return a*a  
x = map(function, (1,2,3,4)) #x is the map object  
print(x)  
print(set(x))
```

Output

{16, 1, 4, 9}

The filter() function is used to generate an output list of values that return true when the function is called.

Code:

```
def func(x):  
    if x>=3:  
        return x  
y = filter(func, (1,2,3,4))  
print(y)  
print(list(y))
```

Output

[3, 4]